



Mostafa Kermani Nia

✉ mkermani1383@gmail.com | 🌐 mostafa-kermaninia.github.io | 📷 mostafa-kermaninia | 📄 mostafakermaninia

Education

School of Electrical and Computer Engineering, University of Tehran

Tehran, Iran

B.SC. IN COMPUTER ENGINEERING

Sept 2022 – Present

- GPA: 19.76/20

National Organization for Development of Exceptional Talents (NODET)

Karaj, Iran

DIPLOMA IN MATHEMATICS AND PHYSICS

Sept 2018 – May 2022

- GPA: 19.90/20

Academic & Teaching Experience

Data Science (CHIEF TEACHING ASSISTANT)

Jan 2026 – Present

- Leading 20+ TAs for 100+ students; refocused curriculum on Big Data (Spark, Kafka) & MLOps.
- Designing modules on Distributed Computing and scalable model deployment using Kubernetes/Docker.
- Integrating Agentic AI, LLMs, and Probabilistic Models (MCMC). Sup.: Dr. Y. Yaghoobzadeh & Dr. S. F. Razavi.

Machine Learning (TEACHING ASSISTANT IN JOINT UGRAD/GRAD COURSES)

Sept 2024 – Jan 2026

- Final Project Mentor; Designed SVM (Kernel methods) assignment. Sup.: Prof. Araabi & Dr. Abolghasemi. & Dr. Tavassolipour.

Engineering Probability and Statistics (CHIEF TEACHING ASSISTANT)

July 2025 – Jan 2026

- Managed 33 TAs for 180+ students; Established **Official GitHub Repo**. Sup.: Prof. A. Vahabie. & Prof. Behnam Bahrak.

Computer Vision (TEACHING ASSISTANT)

Jan 2026 – Present

- Assisting in Image Processing assignments under the supervision of Dr. M. Seifipour.

Operating Systems (OS) (LAB ASSISTANT & TEACHING ASSISTANT)

Sept 2025 – Present

- Designed XV6 kernel modules for Fine-Grained Locking and Per-CPU variables to optimize concurrency.
- Developed assignments on Priority-Aware Scheduling. Sup.: Prof. Mehdi Kargahi.

Database Systems (TEACHING ASSISTANT)

Jan 2026 – Present

- Designed advanced assignments on Relational Database Theory, Schema Normalization (up to BCNF & 4NF), and Lossless Decomposition.
- Sup.: Dr. Azadeh Shakery.

Foundational Computer Science Courses (TEACHING ASSISTANT & LABORATORY

Sept 2023 – Present

ASSISTANT)

- **Computer Architecture Laboratory** (Jan 2026 – Present) – Sup.: Dr. Saeed Safari.
- **Compiler Design & Programming Languages** (Sept 2025 – Present) – Sup.: Dr. F. Ghassemi.
- **Machines and Languages Theory** (Sept 2024 – Jan 2026) – Sup.: Dr. H. Hojjat & Dr. H. Mousavi.
- **Introduction to Computing Systems** (Sept 2023 – Feb 2025) – Sup.: Prof. H. Moradi & Dr. M. R. Hashemi.
- **Discrete Mathematics** (Jan 2024 – July 2024) – Sup.: Prof. S. Mohammadi.
- **Fundamentals of Programming** (Jan 2024 – July 2024) – Sup.: Dr. M. J. Dousti.

ACM Student Chapter (CORE MEMBER)

July 2024 – Sept 2025

- Organized ICPC contests and technical workshops.

Professional Experience

Grantia (AI-Powered Grant Matching)

Remote (Sweden), Pikan Ventures

PRINCIPAL ENGINEER & TECH LEAD

Contract

- Architected an end-to-end RAG Pipeline using Vector Search and LLMs (Gemini); Engineered a Python ETL Infrastructure to normalize fragmented data.
- Developed a high-performance full-stack solution (React, Node.js, Supabase) with optimized SQL queries, reducing latency for real-time visualization.

Majara (Event Management Ecosystem)

Remote (Sweden), Pikan Ventures

PRINCIPAL ENGINEER & TECH LEAD

Contract

- Designed a scalable Event-Driven Architecture (Async Python) handling high concurrency; Integrated a GenAI pipeline for automated asset creation.
- Implemented enterprise-grade Observability (Distributed Tracing/Logging) to ensure system reliability and rapid incident response.

MOMIS (Secure Gaming Platform)

Remote (Finland), PomeGroup

LEAD SOFTWARE ARCHITECT & CO-FOUNDER

April 2025 – Present

- Engineered a Zero-Trust game engine (60 FPS) with server-side Heuristic Analysis to detect non-human patterns and neutralize bot attacks.
- Architected a centralized state-management system to ensure real-time synchronization, minimizing latency in competitive gameplay.

ONTON (Web3 Community Layer)

Remote (Finland), PomeGroup

BLOCKCHAIN INFRASTRUCTURE LEAD

Nov 2025 – Present

- Engineered Trustless Payment Verification via TON SDK and secure wallet authentication using Cryptographic Signatures to prevent replay attacks.
- Designed a scalable event listener to index smart contract interactions, ensuring data consistency for high-volume traffic (900K+ users).

Honors

- Top Student:** Ranked 1st among all Computer Engineering B.Sc students of the 2022 entry cohort, maintaining the top rank for 6 consecutive semesters.
- National University Entrance Exam:** Ranked in the Top 0.6% in Nation-wide Iranian University Entrance Exam (Summer 2022).
- Physics Olympiad:** Silver medal at Iranian National Olympiad in Physics (Summer 2021).

Research Interests

- Systems for Machine Learning:** Memory-efficient LLM fine-tuning, hardware-aware deep learning, and scalable distributed AI architectures.
- Applied ML & Real-World Data Modeling:** Spatio-temporal forecasting, federated learning on decentralized data, and predictive modeling for healthcare applications.
- Trustworthy AI & Scalable Data Engineering:** Reliable data pipelines, domain generalization, and interpretable models for critical decision-making.

Skills

Programming Languages	Python, C/C++, Java, JavaScript, SQL, MATLAB, Verilog
Distributed Systems & Databases	HBase, ClickHouse, PostgreSQL, MongoDB, Elasticsearch, Memcached, PuppyGraph
Web Development & DevOps	Next.js, HTML/CSS, Supabase, Prometheus, Git, Trello, TeX , ...
Machine Learning Libraries	TensorFlow, scikit-learn, Pandas, NumPy, Matplotlib, ...

Projects

Audio Analysis Research Framework Developed a modular research pipeline for Speaker ID, implementing Spectral Subtraction for noise reduction. Authored a technical report evaluating SVM vs. Deep Neural Networks on prosodic features.	Python, Librosa, TF (GitHub)
Automated Data Engineering Pipeline Designed an end-to-end ETL pipeline using Dockerized MySQL and GitHub Actions automation. Implemented complex SQL queries for feature extraction, ensuring data consistency across distributed systems.	Docker, SQL, CI/CD (GitHub)
XV6 System Projects [P1] [P2] [P3] [P4] [P5] Implemented core OS subsystems focusing on Memory Management , Fine-Grained Locking, and Scheduling.	C, x86 Assembly
CPY Compiler Toolkit & Static Analyzer Built a modular compiler infrastructure featuring a Source-to-Source Transpiler (C to CPY) and a custom Static Analysis Engine to detect memory leaks and scope vulnerabilities during compilation.	Java, ANTLR4 (GitHub)
Real-Time Network Telemetry Engine Architected a programmable data plane using P4 to implement In-Band Network Telemetry (INT). Engineered stateful analytics with P4 Registers to detect micro-bursts and queue latency in real-time.	P4, Python, Mininet (GitHub)

Interests

Classical Guitar (11 years of professional training & performance) · Mountaineering & Nature Trekking

Languages

Persian Native
English Upper-intermediate proficiency